

Introduction to Cryptography

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Computational indistinguishability

- Fix a scheme Π and some adversary $\mathcal{A}$
- Define a randomized exp't $\text{PrivK}_{\mathcal{A},\Pi}(n)$:
 - $\mathcal{A}(1^n)$ outputs $m_0, m_1 \in \{0, 1\}^*$ of equal length
 - $k \leftarrow \text{Gen}(1^n)$, $b \leftarrow \{0, 1\}$, $c \leftarrow \text{Enc}_k(m_b)$
 - $b' \leftarrow \mathcal{A}(c)$
 - Adversary $\mathcal{A}$ *succeeds* if $b = b'$, and we say the experiment evaluates to 1 in this case

Computational indistinguishability

- Π is *computationally indistinguishable* if for all PPT attackers $\mathcal{A}$, there is a negligible function ϵ such that $\Pr[\text{PrivK}_{\mathcal{A},\Pi}(n) = 1] \leq \frac{1}{2} + \epsilon(n)$

Warm-up: is this scheme indistinguishable?

- Consider encryption scheme Π that encrypts a $2n$ -bit message using an n -bit key via $\text{Enc}_k(m_a \parallel m_b) = k \oplus m_a \parallel k \oplus m_b$. Which of the following could be the start of a proof that Π is not computationally indistinguishable?
 - Consider an attacker $\mathcal{A}$ who outputs $m_0 = 0^n 0^n$ and $m_1 = 1^n 1^n \dots$
 - Consider an attacker $\mathcal{A}$ who outputs $m_0 = 0^n 0^n$ and $m_1 = 0^n 1^n \dots$
 - Consider an attacker $\mathcal{A}$ who outputs $m_0 = 0^n 1^n$ and $m_1 = 1^n 0^n \dots$
 - Π is computationally indistinguishable, since it uses the one-time pad

Multiple-message Security and CPA

Multiple-message security

- Fix $\Pi, \mathcal{A}$
- Define a randomized exp't $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n)$:
 - $\mathcal{A}(1^n)$ outputs two **vectors** $(m_{0,1}, \dots, m_{0,t})$ and $(m_{1,1}, \dots, m_{1,t})$
 - * Require that $|m_{0,i}| = |m_{1,i}|$ for all i
 - $k \leftarrow \text{Gen}(1^n), \quad b \leftarrow \{0, 1\}, \quad \text{for all } i: c_i \leftarrow \text{Enc}_k(m_{b,i})$
 - $b' \leftarrow \mathcal{A}(c_1, \dots, c_t); \quad \mathcal{A} \text{ succeeds if } b = b', \text{ and experiment evaluates to 1 in this case}$

A formal definition

- Π is *multiple-message indistinguishable* if for all PPT attackers $\mathcal{A}$, there is a negligible function ϵ such that $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n) = 1] \leq \frac{1}{2} + \epsilon(n)$
- Exercise: show that the pseudo-OTP is *not* multiple-message indistinguishable

Multiple-message secrecy

- No encryption scheme is multiple-message indistinguishable!
 - Proof?
- What assumption did we make?
- No **deterministic** (and stateless) encryption scheme is multiple-message indistinguishable
 - Proof sketch on next slide
 - Need to consider **randomized** schemes!

Proof: repeated-message attack

- Assume **deterministic, stateless** encryption with perfect correctness.
- Choose distinct, equal-length messages m, m' and submit

$$\mathbf{m}_0 = (m, m), \quad \mathbf{m}_1 = (m, m').$$

- Given (c_1, c_2) , output $b' = 0$ if $c_1 = c_2$; otherwise output $b' = 1$.
- If $b = 0$: the same message and key give $c_1 = c_2$ by determinism.
- If $b = 1$: correctness gives $c_1 \neq c_2$; one ciphertext cannot decrypt to both m and m' .
- This efficient attacker wins with probability 1, violating the bound $\frac{1}{2} + \text{negl}(n)$.

Randomized encryption

- The issue is *not* an artifact of our definition
 - It really is a problem if an attacker can tell when the same message is encrypted twice

Multiple-message secrecy

- We are not going to work with multiple-message indistinguishability
- Instead, define something *stronger*: security against chosen-plaintext attacks (CPA-security)
 - Nowadays, this is the *minimal* notion of security an encryption scheme should satisfy

CPA-security

Attacker $\mathcal{A}$

Encryption oracle

Secret key k

Chosen plaintext m_1

$c_1 \leftarrow \text{Enc}_k(m_1)$

Chosen plaintext m_2 (may depend on c_1)

$c_2 \leftarrow \text{Enc}_k(m_2)$

The same secret key is used for every query.

Is the threat model too strong?

- In practice, there are many ways an attacker can *influence* what gets encrypted
 - Not clear how best to model this
 - Chosen-plaintext attacks encompass any such influence
- In some cases an attacker may have complete control over what gets encrypted

“Midway”

- 1 Japanese message: “Will attack AF”
Which location does AF identify?
↓
- 2 Midway broadcasts: “Help! Fresh water needed”
↓
- 3 Japanese message: “AF is out of water”
U.S. analysts identify AF as Midway.

For more details: <http://www.navy.mil/midway/how.html>

Crypto lesson

- “AF” is a plaintext message picked by U.S. analysts, and the Japanese message is a ciphertext that they can influence.

CPA-security

- Fix Π , $\mathcal{A}$
- Define a randomized exp't $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n)$:
 - $k \leftarrow \text{Gen}(1^n)$
 - $\mathcal{A}(1^n)$ interacts with an *encryption oracle* $\text{Enc}_k(\cdot)$, and then outputs m_0, m_1 of the same length
 - $b \leftarrow \{0, 1\}$, $c \leftarrow \text{Enc}_k(m_b)$, give c to $\mathcal{A}$
 - $\mathcal{A}$ can continue to interact with $\text{Enc}_k(\cdot)$
 - $\mathcal{A}$ outputs b'
- $\mathcal{A}$ *succeeds* if $b = b'$, and experiment evaluates to 1 in this case

CPA-security

- Π is *secure against chosen-plaintext attacks (CPA-secure)* if for all PPT attackers $\mathcal{A}$, there is a negligible function ϵ such that $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \epsilon(n)$

Relation with previous definitions?

- CPA-security is stronger than multiple-message indistinguishability
 - **Stronger** means it considers a more powerful adversary
- I.e., if Π is CPA-secure then it is also multiple-message indistinguishable
 - Corollary: no **deterministic** encryption scheme can be CPA-secure

Pseudorandom Functions

Random function

- When we talk about a random function f , we mean
 - Choosing f uniformly at random (and then fixing it)
 - Interacting with f
- In particular, once we choose f there is no more randomness involved
 - In particular, if we query f on the same input twice, we get the same result

Choosing a uniform function

Example: $n = 3$

x	$f(x)$
000	010
001	100
010	100
011	111
100	001
101	010
110	010
111	000

$2^3 = 8$ entries

Func_n : all functions mapping $\{0, 1\}^n$ to $\{0, 1\}^n$.

- 2^n inputs, with an n -bit output for each
- A function table contains $n \cdot 2^n$ bits
- Each table defines one function

$$|\text{Func}_n| = (2^n)^{2^n} = 2^{n \cdot 2^n}$$

Quick check: counting number of functions

- How many functions are there mapping $\{0, 1\}^n$ to $\{0, 1\}^m$?
 - $m \cdot 2^n$
 - $2^{n \cdot 2^m}$
 - $m \cdot 2^{n \cdot 2^n}$
 - $2^{m \cdot 2^n}$

Choosing a uniform function

- Choose uniform $f \in \text{Func}_n$
- Equivalent: for each $x \in \{0, 1\}^n$, choose $f(x)$ uniformly in $\{0, 1\}^n$
 - I.e., fill up the function table with uniform values
 - Can also view this as being done “on-the-fly,” as values are needed

Pseudorandom functions

- Informally, a pseudorandom function “looks like” a random (i.e., uniform) function

Pseudorandom functions

- Informally, a pseudorandom function “looks like” a random function
- As in our discussion of PRGs, it does not make sense to talk about any *fixed* function being pseudorandom
 - We look instead at functions chosen according to some distribution
 - In particular, we look instead at *keyed* functions

Keyed functions

- Let $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ be an efficient, deterministic algorithm
 - Define $F_k(x) = F(k, x)$
 - The first input is called the *key*
- Assume F is *length preserving*: $F(k, x)$ only defined if $|k| = |x|$, in which case $|F(k, x)| = |k| = |x|$
- Choosing a uniform $k \in \{0, 1\}^n$ is equivalent to choosing the function $F_k : \{0, 1\}^n \rightarrow \{0, 1\}^n$
 - The algorithm F defines a *distribution* over functions in Func_n !

Note

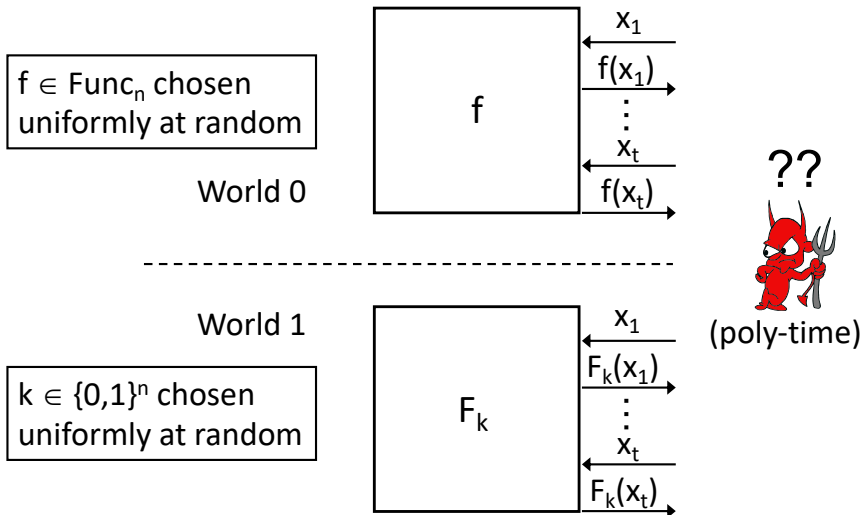
- The number of functions in Func_n is $2^{n \cdot 2^n}$
- $\{F_k\}_{k \in \{0,1\}^n}$ is a subset of Func_n
 - The number of functions in $\{F_k\}_{k \in \{0,1\}^n}$ is at most 2^n
 - This is only a tiny fraction of Func_n !

Pseudorandom functions (PRFs)

- F is a *pseudorandom function* if F_k , for uniform key $k \in \{0, 1\}^n$, is indistinguishable from a uniform function $f \in \text{Func}_n$
- Formally, for all poly-time distinguishers D :

$$\left| \Pr_{k \leftarrow \{0,1\}^n} [D^{F_k(\cdot)} = 1] - \Pr_{f \leftarrow \text{Func}_n} [D^{f(\cdot)} = 1] \right| \leq \epsilon(n)$$

PRF security: the two worlds



Examples (insecure)

- $F(k, x) = 0^n$
- $F(k, x) = k$
- $F(k, x) = k \oplus x$